Week 4: Transformations and Expectations

Week 4: Transformations and Expectations

Reading: Casella & Berger (C&B) §§2.1–2.2

Lecture	Date	Topics	C&B
4.1	Tue. Oct. 20	Distributions of functions of a random variable: the discrete case, the cdf method, monotone transformations, piecewise-monotone transformations	§2.1
4.2	Thu. Oct. 22	The probability integral transformation; expected values; linearity and properties of expectation; $\mathrm{E}X$ as the best predictor	§§2.1–2.2

> "We want something more than mere theory and preaching now, though." (Sherlock Holmes, *A Study in Scarlet*; the epigraph of C&B Chapter 2)

Notation. $Y = g(X)$ is a function of the random variable X. $\mathcal{X}$ and $\mathcal{Y}$ are the support sets of X and Y, as in (2.1.7). g^{-1} is the inverse mapping, defined on sets in (2.1.1). $\mathrm{E}\, g(X)$ is the expected value of $g(X)$.

Where we are. In Week 3 we learned to describe a single random variable X by its cdf F_X and its pmf or pdf f_X. This week answers two natural follow-up questions:

1. If we know the distribution of X, what is the distribution of $Y = g(X)$?
2. What single number summarizes the "center" of a distribution?

Distributions of Functions of a Random Variable

> **Way of thinking.** Don't transform the density; transform the *event*.

1. The inverse mapping and the discrete case

If X is a random variable, then any function $g(X)$ is also a random variable (it is still a function from S into the real numbers). Writing $y = g(x)$, the function $g(x)$ defines a mapping from the sample space $\mathcal{X}$ of X to a new sample space $\mathcal{Y}$, the sample space of Y:

$$g(x) : \mathcal{X} \to \mathcal{Y}.$$

We associate with g an inverse mapping g^{-1}, which maps subsets of $\mathcal{Y}$ to subsets of $\mathcal{X}$:

$$g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}. \tag{2.1.1}$$

g^{-1} takes **sets into sets**, and it exists even when g is not one-to-one. For a point set we write $g^{-1}(y) = \{x \in \mathcal{X} : g(x) = y\}$. This can contain several points; if it contains exactly one point x, we write $g^{-1}(y) = x$.

The distribution of $Y = g(X)$ is then determined by that of X: for any set $A \subset \mathcal{Y}$,

$$P(Y \in A) = P(g(X) \in A) = P\big(\{x \in \mathcal{X} : g(x) \in A\}\big) = P\big(X \in g^{-1}(A)\big). \tag{2.1.2}$$

This is the same idea as the induced probability (1.4.2) from Week 3, applied one level up.

Discrete case. If X is discrete, then $\mathcal{X}$ is countable, so $\mathcal{Y} = \{y : y = g(x),\ x \in \mathcal{X}\}$ is countable and Y is discrete. From (2.1.2), the pmf of Y is

$$f_Y(y) = P(Y = y) = \sum_{x \in g^{-1}(y)} P(X = x) = \sum_{x \in g^{-1}(y)} f_X(x), \quad \text{for } y \in \mathcal{Y},$$

and $f_Y(y) = 0$ for $y \notin \mathcal{Y}$. So: identify $g^{-1}(y)$ for each $y \in \mathcal{Y}$, and add up the probabilities.

Example 2.1.1 (Binomial transformation). A discrete random variable X has a **binomial distribution** if its pmf is of the form

$$f_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n, \tag{2.1.3}$$

where n is a positive integer and $0 \le p \le 1$. Values such as n and p that can be set to different values, producing different probability distributions, are called **parameters**. Consider $Y = g(X)$ with $g(x) = n - x$, so $Y = n - X$. Here $\mathcal{X} = \mathcal{Y} = \{0, 1, \ldots, n\}$, and $g^{-1}(y)$ is the single point $x = n - y$. So

$$f_Y(y) = f_X(n-y) = \binom{n}{n-y} p^{n-y}(1-p)^y = \binom{n}{y}(1-p)^y p^{n-y},$$

using $\binom{n}{y} = \binom{n}{n-y}$ (Definition 1.2.17). Thus Y is also binomial, with parameters n and $1-p$. (If X counts heads, Y counts tails.)

2. The cdf method

For continuous X, the cdf of $Y = g(X)$ is

$$F_Y(y) = P(Y \le y) = P(g(X) \le y) = P\big(\{x \in \mathcal{X} : g(x) \le y\}\big) = \int_{\{x \in \mathcal{X}:\, g(x) \le y\}} f_X(x)\, dx. \quad (2.1.4)$$

The hard part is identifying the set $\{x \in \mathcal{X} : g(x) \le y\}$.

Example 2.1.2 (Uniform transformation). Suppose X has a uniform distribution on $(0, 2\pi)$, that is, $f_X(x) = 1/(2\pi)$ for $0 < x < 2\pi$ and 0 otherwise. Consider $Y = \sin^2(X)$. For $0 < y < 1$, the line at height y crosses $\sin^2(x)$ at four points $x_1 < x_2 < \pi < x_3 < x_4$ (C&B Figure 2.1.1), and

$$P(Y \le y) = P(X \le x_1) + P(x_2 \le X \le x_3) + P(X \ge x_4). \quad (2.1.5)$$

By the symmetry of $\sin^2(x)$ and the uniformity of X,

$$P(Y \le y) = 2P(X \le x_1) + 2P(x_2 \le X \le \pi), \quad (2.1.6)$$

where x_1 and x_2 are the two solutions to $\sin^2(x) = y$, $0 < x < \pi$. Even this simple-looking transformation leads to a cdf that takes some work. (Carrying it through with $x_1 = \arcsin\sqrt{y}$ and $x_2 = \pi - x_1$ gives $F_Y(y) = \frac{2}{\pi}\arcsin\sqrt{y}$ for $0 < y < 1$; see Exercise 2.5.)

Support sets. To avoid confusion, always keep track of where the random variables live. For $Y = g(X)$ it is most convenient to use

$$\mathcal{X} = \{x : f_X(x) > 0\} \quad \text{and} \mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}. \quad (2.1.7)$$

The set where a pdf (or pmf, or any nonnegative function) is positive is called its **support set**, or simply its **support**.

3. Monotone transformations

g is **monotone** if it is either increasing ($u > v \Rightarrow g(u) > g(v)$) or decreasing ($u < v \Rightarrow g(u) > g(v)$). A monotone g is one-to-one and onto from $\mathcal{X}$ to $\mathcal{Y}$, so g^{-1} is single-valued: $g^{-1}(y) = x$ if and only if $y = g(x)$. Then

$$\{x \in \mathcal{X} : g(x) \le y\} = \{x \in \mathcal{X} : x \le g^{-1}(y)\} \quad \text{if } g \text{ is increasing,} \quad (2.1.8)$$

$$\{x \in \mathcal{X} : g(x) \le y\} = \{x \in \mathcal{X} : x \ge g^{-1}(y)\} \quad \text{if } g \text{ is decreasing.} \quad (2.1.9)$$

(A decreasing function reverses inequalities: draw a graph to see why.)

Theorem 2.1.3. Let X have cdf $F_X(x)$, let $Y = g(X)$, and let $\mathcal{X}$ and $\mathcal{Y}$ be defined as in (2.1.7).

- **a.** If g is an increasing function on $\mathcal{X}$, $F_Y(y) = F_X\big(g^{-1}(y)\big)$ for $y \in \mathcal{Y}$.
- **b.** If g is a decreasing function on $\mathcal{X}$ and X is a continuous random variable, $F_Y(y) = 1 - F_X\big(g^{-1}(y)\big)$ for $y \in \mathcal{Y}$.

Why. For increasing g, by (2.1.8), $F_Y(y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y))$. For decreasing g, by (2.1.9), $F_Y(y) = P(X \ge g^{-1}(y)) = 1 - F_X(g^{-1}(y))$, where the last step uses continuity: $P(X = g^{-1}(y)) = 0$.

Example 2.1.4 (Uniform-exponential relationship–I). Suppose $X \sim f_X(x) = 1$ if $0 < x < 1$ and 0 otherwise, the **uniform(0, 1) distribution**, so $F_X(x) = x$ for $0 < x < 1$. Let $Y = g(X) = -\log X$.

Since

$$\frac{d}{dx}g(x) = \frac{d}{dx}(-\log x) = -\frac{1}{x} < 0, \quad \text{for } 0 < x < 1,$$

g is decreasing, and $\mathcal{Y} = (0, \infty)$. For $y > 0$, $y = -\log x$ implies $x = e^{-y}$, so $g^{-1}(y) = e^{-y}$ and

$$F_Y(y) = 1 - F_X\big(g^{-1}(y)\big) = 1 - F_X(e^{-y}) = 1 - e^{-y}.$$

Of course $F_Y(y) = 0$ for $y \le 0$. Note that we only needed g to be monotone on $(0, 1)$, the support of X.

Theorem 2.1.5. Let X have pdf $f_X(x)$ and let $Y = g(X)$, where g is a monotone function. Let $\mathcal{X}$ and $\mathcal{Y}$ be defined by (2.1.7). Suppose that $f_X(x)$ is continuous on $\mathcal{X}$ and that $g^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$. Then the pdf of Y is given by

$$f_Y(y) = \begin{cases} f_X\big(g^{-1}(y)\big)\left|\dfrac{d}{dy}g^{-1}(y)\right| & y \in \mathcal{Y} \\ 0 & \text{otherwise.} \end{cases} \tag{2.1.10}$$

Proof. From Theorem 2.1.3, by the chain rule,

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \begin{cases} f_X\big(g^{-1}(y)\big)\dfrac{d}{dy}g^{-1}(y) & \text{if } g \text{ is increasing} \\ -f_X\big(g^{-1}(y)\big)\dfrac{d}{dy}g^{-1}(y) & \text{if } g \text{ is decreasing,} \end{cases}$$

which can be expressed concisely as (2.1.10). □

Why the absolute value. $f_X(g^{-1}(y))$ says *where* y comes from. The factor $|dx/dy|$ says how much g stretches or squeezes a small interval: probability is conserved, $f_Y(y)\,|dy| = f_X(x)\,|dx|$, and densities are never negative.

Example 2.1.6 (Inverted gamma pdf). Let $f_X(x)$ be the **gamma pdf**

$$f(x) = \frac{1}{(n-1)!\,\beta^n}x^{n-1}e^{-x/\beta}, \quad 0 < x < \infty,$$

where β is a positive constant and n is a positive integer. To find the pdf of $g(X) = 1/X$, note that $\mathcal{X} = \mathcal{Y} = (0, \infty)$, $g^{-1}(y) = 1/y$, and $\frac{d}{dy}g^{-1}(y) = -1/y^2$. By Theorem 2.1.5, for $y \in (0, \infty)$,

$$\begin{aligned} f_Y(y) &= f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy}g^{-1}(y)\right| = \frac{1}{(n-1)!\,\beta^n}\left(\frac{1}{y}\right)^{n-1}e^{-1/(\beta y)}\frac{1}{y^2} \\ &= \frac{1}{(n-1)!\,\beta^n}\left(\frac{1}{y}\right)^{n+1}e^{-1/(\beta y)}, \end{aligned}$$

a special case of the **inverted gamma pdf**.

4. Piecewise-monotone transformations

Often g is neither increasing nor decreasing, but it is monotone on pieces. (If g is not monotone over certain intervals, then, as C&B put it, "we are in *deep* trouble.")

Example 2.1.7 (Square transformation). Suppose X is a continuous random variable. For $y > 0$, the cdf of $Y = X^2$ is

$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y}).$$

Because X is continuous, we can drop the equality from the left endpoint:

$$F_Y(y) = P(-\sqrt{y} < X \le \sqrt{y}) = P(X \le \sqrt{y}) - P(X \le -\sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

Differentiating, with the chain rule for each term,

$$f_Y(y) = \frac{d}{dy}\left[F_X(\sqrt{y}) - F_X(-\sqrt{y})\right] = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}),$$

so

$$f_Y(y) = \frac{1}{2\sqrt{y}}\left(f_X(\sqrt{y}) + f_X(-\sqrt{y})\right). \tag{2.1.11}$$

The pdf of Y is a sum of two pieces, one for each interval on which $g(x) = x^2$ is monotone. In general, this will be the case.

Theorem 2.1.8. Let X have pdf $f_X(x)$, let $Y = g(X)$, and define the sample space $\mathcal{X}$ as in (2.1.7). Suppose there exists a partition, $A_0, A_1, \ldots, A_k$, of $\mathcal{X}$ such that $P(X \in A_0) = 0$ and $f_X(x)$ is continuous on each A_i. Further, suppose there exist functions $g_1(x), \ldots, g_k(x)$, defined on $A_1, \ldots, A_k$, respectively, satisfying

- **i.** $g(x) = g_i(x)$, for $x \in A_i$,
- **ii.** $g_i(x)$ is monotone on A_i,
- **iii.** the set $\mathcal{Y} = \{y : y = g_i(x) \text{ for some } x \in A_i\}$ is the same for each $i = 1, \ldots, k$, and
- **iv.** $g_i^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$, for each $i = 1, \ldots, k$.

Then

$$f_Y(y) = \begin{cases} \displaystyle\sum_{i=1}^{k} f_X\big(g_i^{-1}(y)\big) \left|\frac{d}{dy} g_i^{-1}(y)\right| & y \in \mathcal{Y} \\ 0 & \text{otherwise.} \end{cases}$$

Reading the theorem. Split $\mathcal{X}$ into pieces $A_1, \ldots, A_k$ on which g is monotone. Each g_i is a one-to-one map from A_i onto $\mathcal{Y}$, so each y has exactly one "pre-image" $g_i^{-1}(y)$ in each piece, and $f_Y(y)$ adds up the contributions from all the pieces. The "exceptional set" A_0, which has probability 0, is a technical device for handling endpoints such as $x = 0$ below. (See Exercise 2.7 for an extension.)

Example 2.1.9 (Normal-chi squared relationship). Let X have the **standard normal distribution**,

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty,$$

and let $Y = X^2$. The function $g(x) = x^2$ is monotone on $(-\infty, 0)$ and on $(0, \infty)$, and $\mathcal{Y} = (0, \infty)$. Take

$$A_0 = \{0\}; \qquad A_1 = (-\infty, 0),\ g_1^{-1}(y) = -\sqrt{y}; \qquad A_2 = (0, \infty),\ g_2^{-1}(y) = \sqrt{y}.$$

By Theorem 2.1.8,

$$\begin{aligned} f_Y(y) &= \frac{1}{\sqrt{2\pi}} e^{-(-\sqrt{y})^2/2} \left| -\frac{1}{2\sqrt{y}} \right| + \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{y})^2/2} \left| \frac{1}{2\sqrt{y}} \right| \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-y/2}, \quad 0 < y < \infty. \end{aligned}$$

This is the pdf of a **chi squared random variable with 1 degree of freedom**, which we will meet often.

Check your understanding.

1. Let $X \sim$ uniform(0, 1) and $Y = X^2$. Find F_Y and f_Y in two ways: by the cdf method, and by Theorem 2.1.5. Is f_Y bounded?

   > On $(0, 1)$, $g(x) = x^2$ is increasing, so $F_Y(y) = P(X \le \sqrt{y}) = \sqrt{y}$ for $0 < y < 1$. By Theorem 2.1.5 with $g^{-1}(y) = \sqrt{y}$: $f_Y(y) = 1 \cdot \frac{1}{2\sqrt{y}}$, $0 < y < 1$, which matches $\frac{d}{dy}\sqrt{y}$. It is **unbounded** near $y = 0$, yet still integrates to 1: a pdf is not a probability.

2. Let X have pdf f_X and $Y = aX + b$ with $a \neq 0$. Show that $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$.

   > $g(x) = ax + b$ is monotone (increasing if $a > 0$, decreasing if $a < 0$), with $g^{-1}(y) = (y - b)/a$ and $\frac{d}{dy} g^{-1}(y) = 1/a$. Theorem 2.1.5 gives the formula. This is the basis of the location–scale families in Chapter 3.

3. Let X be the number of heads in 3 tosses of a fair coin and $Y = (X - 1)^2$. Find the pmf of Y.

   > $g^{-1}(0) = \{1\}$, $g^{-1}(1) = \{0, 2\}$, $g^{-1}(4) = \{3\}$. So $f_Y(0) = 3/8$, $f_Y(1) = 1/8 + 3/8 = 1/2$, and $f_Y(4) = 1/8$. Two values of x map to $y = 1$, so their probabilities add.

The Probability Integral Transformation and Expected Values

> **Way of thinking.** The expected value is a probability-weighted average.

5. The probability integral transformation

Theorem 2.1.10 (Probability integral transformation). Let X have continuous cdf $F_X(x)$ and define the random variable Y as $Y = F_X(X)$. Then Y is uniformly distributed on $(0, 1)$, that is, $P(Y \le y) = y$, $0 < y < 1$.

The inverse cdf. If F_X is strictly increasing, then F_X^{-1} is well defined by

$$F_X^{-1}(y) = x \iff F_X(x) = y. \tag{2.1.12}$$

But if F_X is constant on some interval $[x_1, x_2]$, then every x in that interval satisfies $F_X(x) = y$ (C&B Figure 2.1.2b), and (2.1.12) does not define a single value. This is avoided by defining, for $0 < y < 1$,

$$F_X^{-1}(y) = \inf\{x : F_X(x) \ge y\}, \tag{2.1.13}$$

which agrees with (2.1.12) when F_X is nonconstant and is single-valued even when F_X is not strictly increasing. (At the endpoints, $F_X^{-1}(1) = \infty$ if $F_X(x) < 1$ for all x, and, for any F_X, $F_X^{-1}(0) = -\infty$.)

Proof of Theorem 2.1.10. For $Y = F_X(X)$ and $0 < y < 1$,

$$\begin{aligned} P(Y \le y) &= P\big(F_X(X) \le y\big) \\ &= P\big(F_X^{-1}[F_X(X)] \le F_X^{-1}(y)\big) && (F_X^{-1} \text{ is increasing}) \\ &= P\big(X \le F_X^{-1}(y)\big) && (\text{see below}) \\ &= F_X\big(F_X^{-1}(y)\big) && (\text{definition of } F_X) \\ &= y. && (\text{continuity of } F_X) \end{aligned}$$

At the endpoints, $P(Y \le y) = 1$ for $y \ge 1$ and $P(Y \le y) = 0$ for $y \le 0$, so Y is uniform(0, 1).

The third equality deserves attention. If F_X is strictly increasing, $F_X^{-1}(F_X(x)) = x$. If F_X is flat on $[x_1, x_2]$, then $F_X^{-1}(F_X(x)) = x_1$ for any x in that interval, which is not x. But the probability equality still holds, because $P(X \le x) = P(X \le x_1)$ for any $x \in [x_1, x_2]$: the flat part of the cdf has probability 0, $P(x_1 < X \le x) = F_X(x) - F_X(x_1) = 0$. $\square$

Application: simulation. To generate an observation X with cdf F_X, generate a uniform(0, 1) random number V and solve $F_X(x) = V$ for x, that is, set $X = F_X^{-1}(V)$. For example, by Example 2.1.4, $-\log V$ has the exponential distribution with $F(y) = 1 - e^{-y}$. This "inverse cdf method" is how computers sample from many distributions. It also gives a concrete answer to Theorem 1.5.3's claim that every cdf belongs to *some* random variable.

6. Expected values

The expected value, or expectation, of a random variable is its average value, where "average" is weighted according to the probability distribution. It is a measure of center.

Definition 2.2.1. The **expected value** or **mean** of a random variable $g(X)$, denoted by $\mathrm{E}\, g(X)$, is

$$\mathrm{E}\, g(X) = \begin{cases} \displaystyle\int_{-\infty}^{\infty} g(x) f_X(x)\, dx & \text{if } X \text{ is continuous} \\ \displaystyle\sum_{x \in \mathcal{X}} g(x) f_X(x) = \sum_{x \in \mathcal{X}} g(x) P(X = x) & \text{if } X \text{ is discrete,} \end{cases}$$

provided that the integral or sum exists. If $\mathrm{E}|g(X)| = \infty$, we say that $\mathrm{E}\, g(X)$ does not exist.

Notice that we compute $\mathrm{E}\, g(X)$ using the distribution of X, without first finding the distribution of $g(X)$. (Ross 1988 calls this the "law of the unconscious statistician"; C&B "do not find this amusing.")

Example 2.2.2 (Exponential mean). Suppose X has an **exponential(λ) distribution**, with pdf

$$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}, \quad 0 \le x < \infty, \quad \lambda > 0.$$

Then, integrating by parts,

$$\mathrm{E}X = \int_0^{\infty} \frac{1}{\lambda} x e^{-x/\lambda}\, dx = -x e^{-x/\lambda}\Big|_0^{\infty} + \int_0^{\infty} e^{-x/\lambda}\, dx = \int_0^{\infty} e^{-x/\lambda}\, dx = \lambda.$$

Example 2.2.3 (Binomial mean). If $X \sim$ binomial(n, p) as in (2.1.3), then (the $x = 0$ term is 0)

$$\mathrm{E}X = \sum_{x=0}^{n} x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^{n} x \binom{n}{x} p^x (1-p)^{n-x}.$$

Using the identity $x\binom{n}{x} = n\binom{n-1}{x-1}$ and substituting $y = x - 1$,

$$\mathrm{E}X = \sum_{x=1}^{n} n \binom{n-1}{x-1} p^x (1-p)^{n-x} = np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{n-1-y} = np,$$

since the last sum is the total of a binomial($n-1$, p) pmf, which is 1. (Trick worth remembering: factor out constants until what remains is a pmf or pdf that sums or integrates to 1.)

Example 2.2.4 (Cauchy mean). A classic random variable whose expected value does not exist is a **Cauchy random variable**, with pdf

$$f_X(x) = \frac{1}{\pi} \frac{1}{1 + x^2}, \quad -\infty < x < \infty.$$

It is straightforward to check that $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$, but $\mathrm{E}|X| = \infty$. For any positive number M,

$$\int_0^M \frac{x}{1+x^2}\, dx = \frac{\log(1+x^2)}{2}\Big|_0^M = \frac{\log(1+M^2)}{2},$$

so

$$\begin{aligned} \mathrm{E}|X| &= \int_{-\infty}^{\infty} \frac{|x|}{\pi} \frac{1}{1+x^2}\, dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2}\, dx \\ &= \lim_{M \to \infty} \frac{1}{\pi} \log(1+M^2) = \infty, \end{aligned}$$

and $\mathrm{E}X$ does not exist. The density is symmetric about 0, but "the positive and negative parts cancel" is not allowed: both are infinite, and $\infty - \infty$ is undefined.

7. Properties of expectation

Taking expectations is a **linear** operation: for any constants a and b,

$$\mathrm{E}(aX + b) = a\,\mathrm{E}X + b. \tag{2.2.1}$$

For example, if $X \sim \text{binomial}(n, p)$, then $\mathrm{E}(X - np) = \mathrm{E}X - np = 0$.

Theorem 2.2.5. Let X be a random variable and let a, b, and c be constants. Then for any functions $g_1(x)$ and $g_2(x)$ whose expectations exist,

- **a.** $\mathrm{E}\big(ag_1(X) + bg_2(X) + c\big) = a\,\mathrm{E}\,g_1(X) + b\,\mathrm{E}\,g_2(X) + c$.
- **b.** If $g_1(x) \geq 0$ for all x, then $\mathrm{E}\,g_1(X) \geq 0$.
- **c.** If $g_1(x) \geq g_2(x)$ for all x, then $\mathrm{E}\,g_1(X) \geq \mathrm{E}\,g_2(X)$.
- **d.** If $a \leq g_1(x) \leq b$ for all x, then $a \leq \mathrm{E}\,g_1(X) \leq b$.

Proof of (a), continuous case. By the additivity of the integral, and since constants factor out,

$$\begin{aligned}
\mathrm{E}\big(ag_1(X) + bg_2(X) + c\big) &= \int_{-\infty}^{\infty} \big(ag_1(x) + bg_2(x) + c\big) f_X(x)\,dx \\
&= a\int_{-\infty}^{\infty} g_1(x) f_X(x)\,dx + b\int_{-\infty}^{\infty} g_2(x) f_X(x)\,dx + c\int_{-\infty}^{\infty} f_X(x)\,dx,
\end{aligned}$$

which is $a\,\mathrm{E}\,g_1(X) + b\,\mathrm{E}\,g_2(X) + c$. The discrete case and parts (b)–(d) are similar. □

Caution. Linearity does **not** extend to nonlinear functions: in general $\mathrm{E}\,g(X) \neq g(\mathrm{E}X)$. For example, if $X \sim \text{uniform}(0, 1)$, then $\mathrm{E}X^2 = 1/3 \neq (\mathrm{E}X)^2 = 1/4$.

Example 2.2.6 (Minimizing distance). Measure the distance between X and a constant b by $(X - b)^2$. Which b minimizes $\mathrm{E}(X - b)^2$? (Minimizing $(X - b)^2$ itself is useless, since the answer would depend on X.) Adding and subtracting $\mathrm{E}X$,

$$\begin{aligned}
\mathrm{E}(X - b)^2 &= \mathrm{E}(X - \mathrm{E}X + \mathrm{E}X - b)^2 \\
&= \mathrm{E}\big((X - \mathrm{E}X) + (\mathrm{E}X - b)\big)^2 \\
&= \mathrm{E}(X - \mathrm{E}X)^2 + (\mathrm{E}X - b)^2 + 2\,\mathrm{E}\big((X - \mathrm{E}X)(\mathrm{E}X - b)\big).
\end{aligned}$$

The last term is $(\mathrm{E}X - b)\,\mathrm{E}(X - \mathrm{E}X) = (\mathrm{E}X - b)(\mathrm{E}X - \mathrm{E}X) = 0$, because $\mathrm{E}X - b$ is a constant. So

$$\mathrm{E}(X - b)^2 = \mathrm{E}(X - \mathrm{E}X)^2 + (\mathrm{E}X - b)^2. \tag{2.2.2}$$

We cannot control the first term, and the second is ≥ 0 with equality when $b = \mathrm{E}X$. Hence

$$\min_b \mathrm{E}(X - b)^2 = \mathrm{E}(X - \mathrm{E}X)^2. \tag{2.2.3}$$

$\mathrm{E}X$ is the **best constant predictor** of X under squared error. (Exercise 2.19 gives a calculus proof; Exercise 2.18 gives the analogous result for the median under absolute error.) The minimum value,

$\mathrm{E}(X - \mathrm{E}X)^2$, is the variance, which starts next week.

Two ways to compute $\mathrm{E}\, g(X)$**.** Directly from the definition,

$$\mathrm{E}\, g(X) = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx, \tag{2.2.4}$$

or by first finding the pdf f_Y of $Y = g(X)$ (Section 2.1) and then

$$\mathrm{E}\, g(X) = \mathrm{E}Y = \int_{-\infty}^{\infty} y f_Y(y)\, dy. \tag{2.2.5}$$

Example 2.2.7 (Uniform-exponential relationship–II). Let $X \sim$ uniform(0, 1), with $f_X(x) = 1$ if $0 \le x \le 1$ and 0 otherwise, and let $g(X) = -\log X$. By (2.2.4),

$$\mathrm{E}\, g(X) = \mathrm{E}(-\log X) = \int_0^1 -\log x\, dx = \big(x - x \log x\big)\Big|_0^1 = 1.$$

By Example 2.1.4, $Y = -\log X$ has cdf $1 - e^{-y}$ and hence pdf $f_Y(y) = \frac{d}{dy}(1 - e^{-y}) = e^{-y}$, $0 < y < \infty$, the exponential pdf with $\lambda = 1$. So by Example 2.2.2 and (2.2.5), $\mathrm{E}Y = 1$, the same answer.

Check your understanding.

1. Let X have pmf $P(X = -1) = 0.2$, $P(X = 0) = 0.5$, $P(X = 2) = 0.3$. Find $\mathrm{E}X$, $\mathrm{E}X^2$, and $\mathrm{E}(3X - 1)$. Is $\mathrm{E}X^2 = (\mathrm{E}X)^2$?

   > $\mathrm{E}X = -0.2 + 0 + 0.6 = 0.4$. $\mathrm{E}X^2 = 0.2 + 0 + 1.2 = 1.4$. $\mathrm{E}(3X - 1) = 3(0.4) - 1 = 0.2$ by (2.2.1). $\mathrm{E}X^2 = 1.4 \neq 0.16 = (\mathrm{E}X)^2$.

2. Let X be continuous with cdf F_X. What is the distribution of $1 - F_X(X)$? Use this to explain why $\mathrm{E}F_X(X) = 1/2$.

   > By Theorem 2.1.10, $U = F_X(X) \sim$ uniform(0, 1). Then $1 - U$ is also uniform(0, 1) (Check your understanding 2 of Lecture 4.1 with $a = -1$, $b = 1$). And $\mathrm{E}F_X(X) = \mathrm{E}U = \int_0^1 u\, du = 1/2$.

3. Does the expected value of the pdf $f(x) = 1/x^2$, $1 < x < \infty$, exist?

   > f is a pdf: $\int_1^\infty x^{-2}\, dx = 1$. But $\mathrm{E}X = \int_1^\infty x \cdot x^{-2}\, dx = \int_1^\infty \frac{dx}{x} = \infty$, so the mean does not exist (here it is $+\infty$, since $X > 0$). Heavy tails can make the mean infinite, as in the Cauchy example.

Summary of Week 4

- **Functions of a random variable:** $P(Y \in A) = P(X \in g^{-1}(A))$ (2.1.2). Discrete case: add $f_X(x)$ over $x \in g^{-1}(y)$.
- **Cdf method:** $F_Y(y) = P(g(X) \le y)$; identify the x-set, then differentiate. Keep track of the supports $\mathcal{X}$ and $\mathcal{Y}$ (2.1.7).
- **Monotone** g**:** $F_Y(y) = F_X(g^{-1}(y))$ or $1 - F_X(g^{-1}(y))$ (Theorem 2.1.3), and $f_Y(y) = f_X(g^{-1}(y)) \left|\frac{d}{dy} g^{-1}(y)\right|$ (Theorem 2.1.5). Piecewise monotone: sum over the pieces

(Theorem 2.1.8).
- **Probability integral transformation:** $F_X(X) \sim$ uniform(0, 1) for continuous F_X; conversely, $F_X^{-1}(V)$ with $V \sim$ uniform(0, 1) has cdf F_X.
- **Expectation:** $\mathrm{E}\, g(X) = \int g(x) f_X(x)\, dx$ or $\sum g(x) f_X(x)$, if $\mathrm{E}|g(X)| < \infty$; it may not exist (Cauchy). It is linear (Theorem 2.2.5), but $\mathrm{E}\, g(X) \neq g(\mathrm{E}X)$ in general. $\mathrm{E}X$ minimizes $\mathrm{E}(X-b)^2$.

Practice Problems

From C&B §2.5:

- **Transformations (discrete and monotone):** 2.1, 2.2, 2.3.
- **Piecewise-monotone transformations:** 2.5, 2.6, 2.7.
- **Inverse cdfs and the probability integral transformation:** 2.8, 2.9, 2.10.
- **Expected values:** 2.11, 2.12, 2.13, 2.14, 2.15, 2.17, 2.18, 2.19, 2.20.

Next week: moments and moment generating functions; differentiating under an integral sign (§§2.3–2.4).